

3. kontrolna naloga

3. A, 5. 2. 2025

Ime in priimek: MITJA ŠEVERKAR

25

dosežene točke	možne točke	odstotki	ocena
38	42	90	5

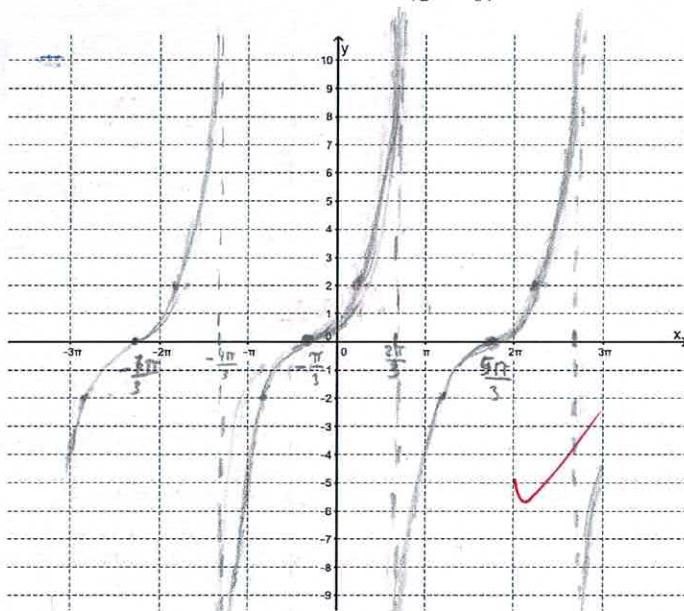
ČAS PISANJA: 45 minut

 $\frac{11}{4}$

1. a) Na intervalu
- $[-3\pi, 3\pi]$
- nariši graf funkcije
- $f(x) = 2 \tan\left(\frac{x}{2} + \frac{\pi}{6}\right)$
- .

[4t] 4

$$= 2 \tan\left(\frac{1}{2}\left(x + \frac{\pi}{3}\right)\right)$$



$$\arctan \sqrt{3} = 60^\circ$$

- b) Izračunaj natančno vrednost izraza (brez uporabe računalnika = viden naj bo računski postopek):
- $\cos(\arccos \frac{1}{3} + \arctan(\sqrt{3}))$
- .

[4t] 2

$$\cos(\arccos \frac{1}{3} + \arctan \sqrt{3}) = \cos \arccos \frac{1}{3} \cdot \cos \arctan \sqrt{3} - \sin \arccos \frac{1}{3} \cdot \sin \arctan \sqrt{3}$$

$$\sin \arctan \sqrt{3} = \frac{1}{3} \cdot \cos 60^\circ - \sin \arccos \frac{1}{3} \cdot \sin 60^\circ =$$

$$= \frac{1}{3} \cdot \frac{1}{2} - \sin \arccos \frac{1}{3} \cdot \frac{\sqrt{3}}{2} = \frac{1}{6} - \frac{\sqrt{3}}{2} \sin(\arccos \frac{1}{3})$$

2. Premica p je podana z enačbo $y = ax + 1$. Premica q je pravokotna na premico z enačbo $4x + 2y + 7 = 0$. Določi vrednost $a \in \mathbb{R}$, da bo kot med premicama p in q enak 45° . [7t] 5

$$y = -4x - \frac{7}{2}$$

$$y = -2x - \frac{7}{2}$$



$$y_p = ax + 1$$

$$y_q = \frac{1}{2}x - \frac{7}{2}$$

$$\tan \varphi = \frac{1}{2}$$

$$\varphi = 26,57^\circ$$

$$\angle = 45^\circ \rightarrow \text{arctan } \frac{1}{2}$$

$$a = \tan \angle = \frac{1}{3} = \tan(45^\circ - \arctan \frac{1}{2})$$

$\neq a_2$

3. Reši enačbi:

a) $\cos^2 x + 3 \sin^2 x + \sqrt{3} \sin(2x) = 1$

[5t] 5

$$\cos^2 x + 3 \sin^2 x + 2\sqrt{3} \sin x \cos x = \sin^2 x + \cos^2 x$$

$$2 \sin^2 x + 2\sqrt{3} \sin x \cos x = 0$$

$$/: \cos^2 x$$

$$\sin x \cdot \sin x$$

$$\tan^2 x + \sqrt{3} \tan x = 0$$

$$\tan x (\tan x + \sqrt{3}) = 0$$

$$\frac{\cos x = 0}{\sin x = 1}$$

$$1 + \sqrt{3} \cdot 0 = 0$$

$$1 = 0$$

① $\tan x = 0$

$$x_1 = \arctan 0 + k\pi =$$

$$= k\pi \quad k \in \mathbb{Z}$$

② $\tan x = -\sqrt{3}$

$$x_2 = \arctan(-\sqrt{3}) + k\pi =$$

$$= -\frac{\pi}{3} + k\pi$$

b) $\cos \frac{x}{2} + \cos(3x) + \cos(4x) = 0$

[5t] 5

$$\cos \frac{x}{2} + 2 \cos \frac{7x}{2} \cdot \cos \frac{x}{2} = 0$$

$$\cos \frac{x}{2} \left(1 + 2 \cos \frac{7x}{2} \right) = 0$$

① $\cos \frac{x}{2} = 0$

$$\frac{\pi}{2} + k\pi = \frac{x}{2}$$

$$x_1 = \pi + 2k\pi$$

② $\cos \frac{7x}{2} = -\frac{1}{2}$

$$\arccos\left(-\frac{1}{2}\right) = \frac{7x}{2}$$

$$\arccos \frac{2\pi}{3} + 2k\pi = \frac{7x_2}{2}$$

$$k \in \mathbb{Z} \rightarrow x_2 = \frac{2 \cdot 2 \cdot \pi}{3 \cdot 7} + \frac{4k\pi}{7} = \frac{4\pi}{21} + \frac{4k\pi}{7}$$

$$x_3 = -\frac{4\pi}{21} + \frac{4k\pi}{7}$$

4. Dani sta kompleksni števili $z_1 = -1 + i$ in $z_2 = 1 + \sqrt{3}i$.

a) Zapiši števili z_1 in z_2 v polarni obliki.

[4t] 4

$$|z_1| = \sqrt{1+1} = \sqrt{2} \quad \tan \varphi_1 = \frac{1}{-1} \Rightarrow \varphi_1 = 135^\circ$$

$$|z_2| = \sqrt{4} = 2 \quad \tan \varphi_2 = \frac{\sqrt{3}}{1} \quad \varphi_2 = 60^\circ$$

$$z_1 = \sqrt{2} \operatorname{cis}(135^\circ)$$

$$z_2 = 2 \operatorname{cis}(60^\circ)$$

b) Izračunaj vrednost števil $w = z_2^6 \cdot \bar{z}_1$ in $u = \frac{z_1}{z_2}$.

[4t] 4

$$z_2^6 = 2^6 \operatorname{cis}(6 \cdot 60^\circ) = 64 \operatorname{cis} 360^\circ = 64 \operatorname{cis} 0$$

$$\bar{z}_1 = \sqrt{2} \operatorname{cis}(225^\circ)$$

$$w = 64 \operatorname{cis} 0 \cdot \sqrt{2} \operatorname{cis}(225^\circ) = 64\sqrt{2} \operatorname{cis}(225^\circ)$$

$$= 64\sqrt{2} \cos 225^\circ + 64\sqrt{2} i \sin 225^\circ = 64\sqrt{2} \left(-\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \right) = -64 - 64i$$

$$u = \frac{z_1}{z_2} = \frac{\sqrt{2}}{2} \operatorname{cis}(75^\circ)$$

c) Določi najmanjše pozitivno število n , da bo u^n realno število.

[2t] 2

$$\varphi = n \cdot 75^\circ$$

$$5 \cdot 180^\circ = 12 \cdot 75^\circ$$

$$\sin \varphi = 0$$

$$6 \cdot 180^\circ = n \cdot 75^\circ$$

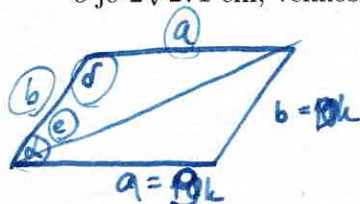
$$n = 12$$

$$n = 12$$

$$n = \frac{12}{5}$$

5. V paralelogramu je razmerje med dolžinama stranic $a : b = 9 : 10$. Dolžina diagonale e je $2\sqrt{271}$ cm, velikost kota α je 60° . Izračunaj ploščino in obseg paralelograma.

[7t] 7



$$\alpha = 60^\circ$$

$$\beta = 120^\circ$$

$$e^2 = a^2 + b^2 - 2ab \cos \beta$$

$$4 \cdot 271 = 81k^2 + 100k^2 - 180k^2 \cos 120^\circ$$

$$1084 = 181k^2 + \frac{1}{2} 180k^2$$

$$1084 = 271k^2$$

$$4 \cdot 271$$

$$k^2 = 4$$

$$k = 2$$

$$a = 9k = 18 \text{ cm}$$

$$b = 10k = 20 \text{ cm}$$

$$o = 2a + 2b = 36 \text{ cm} + 40 \text{ cm} = 76 \text{ cm}$$

$$e = 2\sqrt{271} \text{ cm}$$

$$\sin \alpha = \frac{V_a}{b}$$

$$V_a = \sin \alpha \cdot b = 10\sqrt{3} \text{ cm}$$

~~S =~~

$$S = a \cdot V_a = 18 \text{ cm} \cdot 10\sqrt{3} \text{ cm} = 312 \text{ cm}^2$$

DODATNA NALOGA:

Za katere realne vrednosti a enačba $8 \sin x + a \cos x + 10 = 0$ nima realnih rešitev?

Rešitev naj bo utemeljena.

[3t] 0

$$16 \sin \frac{x}{2} \cdot \cos \frac{x}{2} + a \cos^2 \frac{x}{2} - a \sin^2 \frac{x}{2} + 10 = 0$$

~~*~~

$$D < 0$$

$$/ : \cos \frac{x}{2}$$

$$\frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} = 0$$

$$0 + 0 - a + 10 = 0$$

$$a = -10$$